Hybrid 3D Reconstruction: Combining Classical and Learning-Based Approaches

Shababa Binte Hossain, Jiachen Hua

Abstract

3D reconstruction has advanced through both classical photogrammetry and deep learning. While classical methods like Structure-from-Motion (SfM) and Multi-View Stereo (MVS) provide accurate geometry, they often struggle with realistic textures and lighting, particularly on reflective or non-Lambertian surfaces. In this work, we propose a hybrid pipeline combining the geometric robustness of COLMAP with the rendering capabilities of 3D Gaussian Splatting (3DGS) and 2D Gaussian Splatting (2DGS). We first extract camera poses and sparse/dense representations using COLMAP to train a 3DGS model, subsequently refining texture and appearance with a 2DGS model. Experiments on custom synthetic and real datasets demonstrate that this hybrid approach significantly improves texture quality while maintaining geometric consistency. Quantitative evaluations using LPIPS, total variation metrics, and Cloud-to-Cloud distance (RMS, Mean, Standard Deviation) confirm enhanced perceptual similarity and reduced artifacts, highlighting the effectiveness of integrating classical and learning-based methods for high-quality 3D reconstruction. Further, relighting was implemented on the 3d models for enhanced realism.

I. Introduction

3D reconstruction is a fundamental problem in computer vision with applications in augmented reality, robotics, cultural heritage preservation, and digital content creation. Classical Structure-from-Motion (SfM) and Multi-View Stereo (MVS) pipelines, such as COLMAP[1], are widely used because they provide highly accurate camera pose estimation and geometric reconstruction from multi-view images. However, these methods generally rely on assumptions such as Lambertian reflectance and stable illumination, which limit their performance when scenes contain reflective surfaces, repetitive or deformed textures, shadows, or view-dependent lighting effects.

During our initial experiments with COLMAP on both public and synthetic datasets, we observed that although the reconstructed geometry was often accurate, the resulting textures suffered from visible artifacts. In figure 01, even if the geometry of the reconstruction of the Gerrard Hall (Public Dataset by COLMAP) is appropriate(left), if we zoom into the 3d model, the construction texture and lighting appears unrealistic. Reflective materials such as glass and glossy walls produced “melting” effects, while inconsistent illumination caused unstable texture appearance and degraded visual realism. In addition, reconstruction quality was strongly influenced by image conditions, including lighting direction, texture richness, and feature distinctiveness.

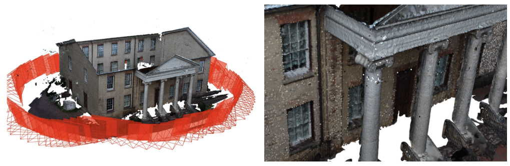


Figure 01: COLMAP reconstruction results

To better analyze these limitations, we incorporated synthetic and real image experiments that allowed controlled evaluation of COLMAP under different illumination and texture conditions. These experiments revealed that reconstruction performance deteriorates when images contain low-texture regions, repetitive patterns, or insufficient multi-directional lighting. Conversely, richer illumination and more distinctive surface features significantly improved feature matching and reconstruction stability.

To address these challenges, we propose a hybrid reconstruction pipeline that combines the geometric robustness of classical methods with the visual advantages of learning-based rendering. Specifically, COLMAP is used to recover reliable camera poses and scene geometry, while 3D Gaussian Splatting (3DGS) & 2D Gaussian

Splatting (2DGS) is employed to improve texture representation and view-dependent appearance modeling. Furthermore, we investigate relighting techniques as a feasibility step toward controllable rendering, enabling reconstructed scenes to be visualized under different lighting conditions. By integrating classical reconstruction, learning-based rendering, synthetic evaluation, and relighting, our approach aims to produce reconstructions that are both geometrically accurate and visually realistic.

II. Background / Related Work

Classical 3D reconstruction pipelines such as COLMAP[1] rely on feature matching, Structure-from-Motion (SfM), and Multi-View Stereo (MVS) to reconstruct scene geometry. These methods are highly effective for geometry but assume Lambertian reflectance, limiting their ability to model complex lighting and texture.

Recent advances in deep learning have introduced neural rendering approaches such as,

- Neural Radiance Fields (NeRF)[13]
- SuGer[11]
- 2DGS[9]
- 3DGS[10]
- Depth Anything V3[12]

We initially attempted to implement NeRF [13]; however, the framework requires a *transforms.json* file, which involves converting COLMAP's *camera.bin* data into a NeRF-compatible format. This additional preprocessing step introduces complexity and increases the overall implementation time. Consequently, we explored alternative approaches that offer more efficient and faster pipelines.

3D Gaussian Splatting (3DGS) is a neural rendering method that represents a scene using millions of semi-transparent 3D Gaussians. Initialized from COLMAP's sparse point cloud and camera poses, it optimizes each Gaussian's position, shape, opacity, and view-dependent color (via spherical harmonics). Through iterative refinement splitting, moving, and pruning Gaussians based on image comparison—it produces highly detailed, photorealistic reconstructions with accurate geometry and rich textures.[10]

2D Gaussian Splatting (2DGS) improves on 3DGS by using 2D planar Gaussians instead of 3D ellipsoids, eliminating volumetric blur and ensuring view-consistent surfaces. Initialized with COLMAP data, it applies perspective-accurate splatting and optimized constraints (depth and normal consistency) to produce sharp textures, precise geometry, and stable, real-time photorealistic reconstructions.[9]

Our work builds on these approaches by combining COLMAP's reliable geometry with 3DGS's advanced appearance modeling. Then improving the results with 2DGS.

Other than these models SuGer[11] is another approach to improve 3DGS output. This method extracts meshes from 3D Gaussian Splatting and enables easy scene composition and animation through mesh manipulation[11]. And Depth Anything V3 that predicts accurate depth maps and generates 3d reconstruction from a single image with strong generalization across diverse scenes[12]. But due to dependency constraints and developer end resource issues, we couldn't effectively implement SuGer[11] and Depth Anything V3[12].

III. Approach

Our proposed pipeline integrates classical and learning-based methods into a unified framework. We made our own synthetic and real dataset and 3D reconstructed it with COLMAP. We post-processed COLMAP outcomes with Cloud Compare. Using COLMAP generated files we applied 3DGS and 2DGS to get improved textured 3D representation. The exported ply file was post processed with Splat editor. To relight the gaussian splat needs to be converted into a mesh file. Which was done with Kiri Engine. Then lighting was implemented on both Synthetic and Real dataset reconstruction.

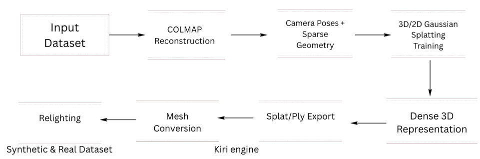


Figure 02: Pipeline for Hybrid 3D reconstruction

IV. Dataset Collection

Our dataset collection process consisted of two components: real-world image acquisition and synthetic image generation. Both were used to evaluate the robustness of the reconstruction pipeline under different texture and lighting conditions.

a.Real Image Dataset

For the real-world dataset, we designed a controlled image acquisition setup to ensure consistent reconstruction conditions. During data collection, the target object was fixed at a stationary position while the illumination conditions were kept constant throughout the capture process. A handheld camera was then moved around the object to record a continuous video from multiple viewpoints, providing sufficient view overlap for reconstruction. Individual frames were later extracted from the video and used as input images for COLMAP.

Figure 03: Real Dataset collection(Reflective Metal, Non reflective plastic, reflective glass jar objects)

We encountered few challenges during image acquisition for COLMAP reconstruction due to insufficient lighting conditions, highly reflective surfaces, uniform/white background, insufficient overlapping. COLMAP relies on Feature-based Multi-View Stereo (SfM), which requires consistent, discriminative, and geometrically reliable pixel correspondences across multiple viewpoints. We found higher image noise in insufficient lighting conditions. COLMAP uses algorithms (like SIFT) to find keypoints[1]. In under-lit areas, the Signal-to-Noise Ratio (SNR) is too low to extract stable features. Shadows move relative to the light source, not the camera. When shadows are present, the software perceives them as physical geometry or surface markings, violating the "static scene" assumption. This causes the matcher to become confused, leading to misalignments or total reconstruction failure for those regions. Shadowed or insufficient lighting caused the results we found in figure 04. We noticed the back or side sections are missing entirely or are extremely sparse. In those areas, the camera likely did not capture enough high-contrast "features" (texture). Shadows effectively "wipe out" the texture of the object, leaving the algorithm with nothing to track.

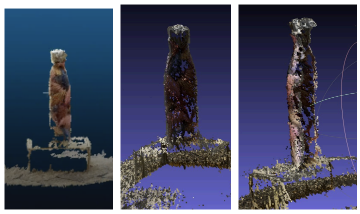


Figure 04: Insufficient lighting caused missing reconstruction on the back part and lower part of the water bottle.

Later on, we selected a reflective jar object as our best final model to conduct our experiment on.

b. Synthetic Image Dataset

To further analyze reconstruction behavior under controlled conditions, we also created synthetic datasets with adjustable textures and illumination settings. The synthetic images allowed us to systematically evaluate how lighting direction, texture richness, and feature distinctiveness influence feature matching and reconstruction stability. Through these experiments, we observed that low-texture and dense repetitive regions often caused reconstruction failures, while multi-directional lighting and stronger surface details improved feature detection and overall reconstruction quality. The synthetic datasets also provided a controlled environment for evaluating relighting and rendering performance in later stages of the pipeline.

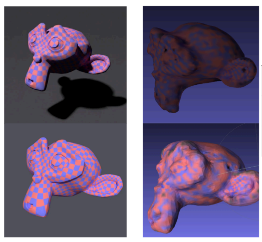


Figure 05: Synthetic data generated and the corresponding 3D reconstruction results

V. COLMAP Reconstruction

After we collected our synthetic and real dataset, we started to model with COLMAP to reconstruct the scene geometry. The pipeline successfully generated both a sparse reconstruction, comprising camera poses and feature points, and a dense reconstruction, which resulted in a detailed point cloud. This process produced several key output files, including images.bin (Extrinsic data), cameras.bin(Intrinsic data), and points3D.bin(Geometry info), which collectively provided the accurate geometric structure and camera calibration necessary for our analysis.

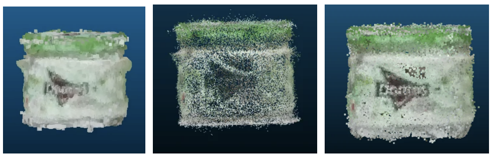


Figure 06: 3D reconstruction on Colmap. a.Poisson mesh, b. COlMAP generated Point clouds, c. Enlarged point clouds (Post processed by Cloud Compare.

After the 3D reconstruction, we post processed the 3D scene with Cloud compare and extracted our 3d model of the object.

VI. Integration with 3D Gaussian Splatting

The COLMAP outputs (sparse file containing images.bin, cameras.bin, points3d.bin, points3d.ply & rigs.bin) were used as input for 3D Gaussian Splatting. The code utilizes these files to map the sparse point cloud into a set of 3D Gaussians, which act as differentiable volumetric primitives. By leveraging the camera poses and intrinsic parameters extracted from the COLMAP pipeline, the model aligns these Gaussians within the scene's coordinate space. We then employed an optimization process gradient descent based on photo-consistency to refine the position, opacity, color, and covariance (shape) of each Gaussian, ultimately enabling the high-fidelity novel view synthesis and dense rendering of the scene. The training process involved optimizing Gaussian primitives to represent both geometry and appearance.

Figure 07: Unprocessed 3D scene generated by 3DGS (7000 Iteration) [Editor: Splat editor]

Following the generated results from 3DGS, we did some postprocessing after each iteration with the online Splat editor to remove the extra environments and to only keep the 3d object. We started with 7000 iterations and recorded results for 10,000, 20,000 and 30,000 iterations. The model converged and produced improved texture quality compared to classical methods like COLMAP.

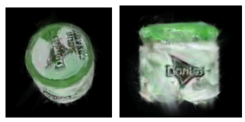
 i. 7000 Iteration (3DGS)	 ii. 10,000 Iteration (3DGS)
 iii. 20,000 Iteration (3DGS)	 iv. 30,000 Iteration (3DGS)

Figure 08: 3DGS results for 7000, 10000, 20000, 30000 iterations

In figure 08, we can notice from the visual outcomes, the 3DGS pipeline result deteriorated after 20,000 iterations.

VII. Integration with 2D Gaussian Splatting (2DGS)

3D Gaussian splitting uses 3D ellipsoids (anisotropic blobs). And for higher iteration these blobs result in foggy and deteriorated texture. To fix the challenges and achieve a cleaner result, we again applied 2D Gaussian Splatting on our COLMAP generated data. 2D Gaussian splatting uses 2D oriented disks (surfels). This time for time constraints we iterated for 7000 and 30,000 iterations.

 I. 2DGS for 7000	

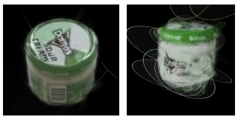
iterations	 Ii. 2DGS for 30,000 iterations

Figure 09: 2DGS results for 7000 and 30000 iterations.

VIII. Relighting Consideration

To analyze lighting limitations, we also explored relighting techniques based on the Lambertian model:

Diffuse color is modeled as: Surface Color = Albedo × Light Intensity × (N · L)

We validated relighting first in 2D before extending it to 3D using surface normals from reconstructed geometry.

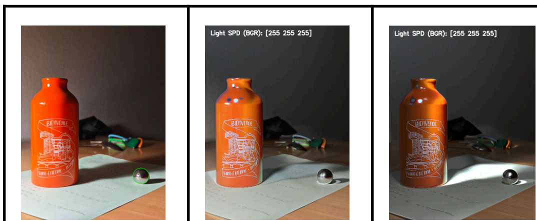


Figure 10: Highlight spot detection for estimating light direction and 2D relighting results

For 3D relighting, the reconstructed 3D mesh was loaded into a rendering environment, where surface normals were obtained directly from the geometry. Virtual light sources with different directions and color temperatures were then applied using the Lambertian reflectance model. This enabled the scene to be visualized under various illumination conditions while preserving its reconstructed geometry.

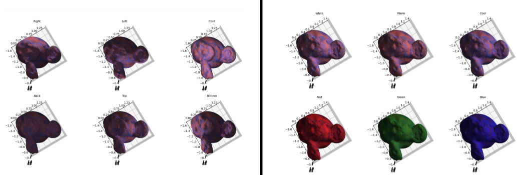


Figure 11: 3D relighting on synthetic images with different light direction and color temperature

Figure 12: 3D relighting on real images with different light direction

IX. Result Analysis

a. Qualitative Results analysis

COLMAP integration with 3DGS reconstruction (Hybrid)

In Figure 08, we noticed Reduced “melting” artifacts in reflective regions then COLMAP results. Improved texture sharpness was noticed for 7000 and 10000 iterations. However,

With increasing iteration the visual outcome deteriorated with the 3DGS pipeline. The deterioration of your 3D Gaussian Splatting (3DGS) resulted at 30,000 iterations is an issue that occurred due to several technical factors in the optimization pipeline. First, the processes of densification and pruning which periodically clone, split, or remove Gaussians can become counterproductive if they continue for too long. By 30k iterations, the model often generates thousands of tiny, "noisy" Gaussians in an attempt to minimize minor remaining pixel errors, resulting in the "hairy" or "fuzzy" cloud visible in the rendered image. This is further exacerbated by Gaussian scaling issues; without strict regularization, individual Gaussians can grow excessively large or thin to cover gaps in the point cloud. While the Gaussians were likely smaller and more concentrated at 7k, by 30k they have "drifted" and stretched, creating a blurry, illusion-like texture instead of a sharp surface.

Compounding this problem is the lack of Opacity Resets in the later stages of training. Standard 3DGS implementations typically reset all opacities near zero every 3,000 iterations to force the model to rebuild and validate them, but this process usually stops after 15,000 iterations. Consequently, between 15k and 30k, low-opacity clutter that should have been deleted is allowed to stay and accumulate, eventually obscuring the actual object.

COLMAP integration with 2DGS reconstruction (Hybrid)

2DGS approach utilizes oriented disks, or surfels. Consequently, we can notice even after 30,000 iterations the results improved better than 3DGS integration. While 3D Gaussian Splatting (3DGS) models objects as a disorganized, volumetric cloud of points that can lose structural coherence over extended training, 2D Gaussian Splatting (2DGS) treats the scene as a structured surface of interlocking planar tiles; this fundamental shift in representation allows 2DGS to maintain superior geometric sharpness and accuracy throughout the optimization process.

b. Gaussian Splat comparison

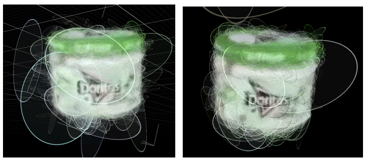


Figure 13: Gaussian splats of 3DGS(Left), Gaussian splats of 2DGS(Right) [Online: Splat Editor]

The primary difference between the Gaussian splats in these two images lies in their fundamental geometry and how they represent the object's surface. In the 3DGS visualization on the left, the primitives are three-dimensional ellipsoids, which appear as rounded, "fat" wireframe blobs. Because these ellipsoids have volume in all three dimensions, they represent the scene as a volumetric cloud or a thick fog. Without a physical constraint to stay flat, these 3D Gaussians tend to "bloat" and overlap in a disorganized manner, creating the "hairy" or fuzzy artifacts seen around the edges of the jar as they attempt to fill space.
On the right, the 2DGS approach utilizes oriented disks, or surfels, which possess zero thickness. These flat primitives are mathematically restricted to tiling the object's surface like a solid shell rather than filling a volume. This surface-oriented constraint, typically reinforced by normal consistency losses, forces the 2D Gaussians to align precisely with the actual geometry of the jar. Because these disks cannot expand into the third dimension, they avoid the "bloating" effect and instead create a clean, refined finish. Ultimately, while 3DGS treats the object as a disorganized collection of 3D blobs that drift apart during long training, 2DGS treats it as a structured surface of interlocking tiles, which is why it remains sharp and accurate even at 30,000 iterations.

c. Overall Qualitative comparison

Figure 14: Visual comparison between Colmap (Left), 3DGS (Middle), 2DGS (Right) mesh results

In figure 14, we can notice the Colmap (Left) result provides the initial foundation but appears fragmented and blocky, as it represents the raw dense point cloud that lacks the continuous color and texture detail needed for photorealism. The 3DGS (Middle) result significantly improves the density and color but introduces substantial "hairy" artifacts and volumetric blur. Because the 3D ellipsoids have too much freedom to expand into empty space, they create a noisy, "bloated" cloud that obscures sharp edges and distorts the jar's labels. Finally, the 2DGS (Right) result delivers the highest quality, providing a clean and sharp mesh. By constraining the Gaussians to flat 2D disks, the model is forced to adhere strictly to the object's surface, resulting in legible text, smooth transitions, and a solid appearance that effectively eliminates the geometric drift and noise seen in the 3DGS version.

X. Quantitative Evaluation

As we got better results from the 2DGS model , for quantitative evaluation we considered comparative analysis among COLMAP and 2DGS.

We used perceptual and texture-based metrics, LPIPS (Learned Perceptual Image Patch Similarity) and Total Variation (Jagginess). LPIPS metric uses a deep learning model to mimic human vision. It doesn't just look at pixels; it looks at "perceptual" quality, such as sharpness, realistic textures, and the absence of artifacts.

Total Variation (TV) calculates the amount of noise or "roughness" in the texture. High values indicate "jagginess," blocky artifacts, or random pixel noise.

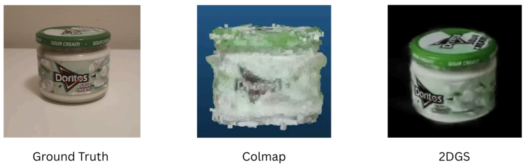


Figure 15: LPIPS and Total Variation analysis among ground truth, Colmap and 2DGS

Table 01: Quantitative analysis among Classical and Hybrid approach

	COLMAP	**Hybrid(2DGS)**
LPIPS	0.8406	0.7681
Total Variation (Jagginess)	7.5600	5.0100

The Hybrid (2DGS) approach outperformed the standard COLMAP baseline across both categories. Specifically, the Hybrid model achieved a lower LPIPS score of 0.7681 (compared to COLMAP's 0.8406), indicating that the final output is perceptually more accurate and exhibits fewer visual distortions to the human eye. Furthermore, the Hybrid method showed a significant reduction in Total Variation, dropping from 7.5600 to 5.0100. This lower 'jagginess' score confirms that the 2DGS constraints successfully filtered out geometric noise and blocky artifacts, resulting in a smoother, more refined surface texture. Collectively, these results provide strong quantitative evidence that the hybrid approach enhances both the visual realism and the structural integrity of the 3D reconstruction.

a. Color Intensity Distribution Analysis

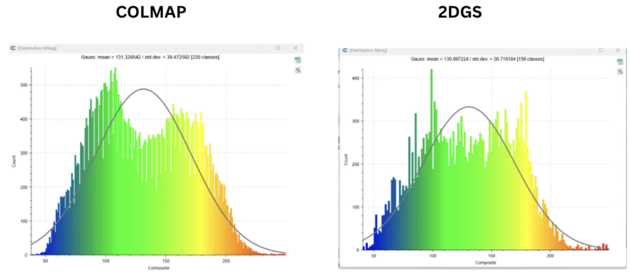


Figure 16: Color Intensity distribution analysis among COLMAP and 2DGS using Cloud Compare

From Colmap, we got mean color intensity of 131.33 with a standard deviation of 39.47. 2DGS histogram tracks the COLMAP distribution very closely, with a nearly identical mean of 130.90 and a slightly narrower standard deviation of 38.72. The high similarity between these distributions proves that 2DGS maintains high color fidelity. It successfully preserves the original dynamic range and lighting of the scene without introducing significant color shifts or artificial oversaturation during the 30k optimization iterations.

Beyond the mean and standard deviation, the reduction in the number of intensity classes from 229 in COLMAP to 156 in 2DGS provides quantitative evidence of noise reduction. While the raw COLMAP data contains a high degree of discrete spectral noise (represented by the higher class count), the 2DGS optimization consolidates these values into a more continuous and stable distribution. This reduction indicates that the 2DGS model has successfully filtered out high-frequency color artifacts and "jitter" found in the initial point cloud, leading to a more visually coherent and smooth surface representation while maintaining the original colorimetric integrity.

b. Geometric Consistency

We evaluated geometric accuracy & consistancy using cloud-to-cloud distance among COLMAP and Hybrid(2DGS) approach:

- RMS Error: 0.0097
- Mean Distance: 0.011043
- Standard Deviation: 0.0106575

To assess the structural fidelity of the reconstruction, a Cloud-to-Cloud Distance analysis was performed, comparing the 2DGS surface to the COLMAP baseline. The Mean Distance of 0.0110 indicates a negligible global geometric error, confirming that the model effectively captures the true scale and shape of the object without the volumetric drift often seen in standard 3DGS. This is supported by an RMS value of 0.0097, which highlights the high precision of the surfel alignment and the absence of significant geometric outliers. Finally, the Standard Deviation of 0.0106 demonstrates excellent geometric consistency across the entire model, proving that the 2DGS constraints successfully produced a uniform, high-fidelity surface that is both mathematically accurate and structurally sound

XII. Conclusion

In this work, we presented a hybrid 3D reconstruction pipeline that combines the strengths of classical and learning-based approaches. By using COLMAP for geometry and 2D Gaussian Splatting for texture modeling, we achieved improved visual quality without compromising structural accuracy.

Our result analysis demonstrated that the hybrid method significantly reduces texture artifacts and improves perceptual realism compared to classical approaches.

To analyze lighting limitations, we also explored relighting techniques based on the Lambertian model. We validated relighting first in 2D before extending it to 3D using surface normals from reconstructed geometry. For 3D relighting, the reconstructed mesh was loaded into a rendering environment, where surface normals were obtained directly from the geometry. Virtual light sources with different directions and color temperatures were then applied using the Lambertian reflectance model. This enabled the scene to be visualized under various illumination conditions while preserving its reconstructed geometry.

XIII. Limitations & Future Work

The experiment can be further improved by implementing the SuGaR pipeline to achieve easier scene composition and animation through mesh manipulation from 3D Gaussian Splatting. The research can also explore a higher number of iterations for both 3DGS and 2DGS. The quantitative comparison was conducted only for 2DGS and COLMAP. The parameters from 3DGS can also be included in this comparison. We could only conduct the hybrid experiment with a single object. Further multi-object scene conditions can be explored.

XIV. References

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